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Project Title: Public Transport Service Frequency and Accessibility in Victoria

Introduction

Public transport services are essential for the ever increasing population of Victoria. About 80% of the population of Victoria use public transport (RACV, n.d.), therefore the accessibility and frequency of public transport services in Victoria is very important. I believe this is a significant topic for citizens, tourists, and government agencies, and would therefore like to find out how well the coverage of the current public transport system is.

Motivation

I have recently moved to Melbourne, Victoria and frequently take public transport as part of my daily commute. I was interested in how frequent and accessible the public transport services were and how this relates to historically higher population growth regions and lower income areas.

Questions

1. Which SA2 areas in Victoria have historically had the greatest population growth, and how well are the current public transport services serving these areas?
2. How does public transport service coverage and frequency vary over different time periods (e.g. time of day/weekends) and how does this service coverage and frequency relate to historically lower income regions?

Data sources

- A. **Population estimates by SA2 and above, 2001 to 2024.** Tabular data in XLSX format with ~2.4K rows x 34 columns. These include SA2, SA3, and SA4 codes and names, state names and population per year recorded. The data was released in March 2025.
(<https://www.abs.gov.au/statistics/people/population/regional-population/2023-24#data-downloads>)
- B. **GTFS Schedule, 2026.** Multiple tabular datasets in TXT format which includes agency, calendar dates, calendar days, levels, pathways, routes, shapes, stop times, stops, transfers, and trips. The data was first published in December 2024 and was last updated March 2026.
(<https://discover.data.vic.gov.au/dataset/gtfs-schedule>)
- C. **ABS - Personal Income - Total Income (SA2) 2011-2018.** Tabular data in XLSX format with ~2.3K rows x 40 columns. These include SA2 codes, SA2 names, earners (persons), median age of earners, mean income, median income, sum of income, SA2 coordinates. The data was published in March 2025.
(<https://data.aurin.org.au/dataset/au-govt-abs-abs-personal-income-total-income-sa2-2011-2018-sa2-2016>)

Data sources A and B will be used to address Question 1, while Question 2 will be answered by using data sources B and C.

References

RACV (n.d.). Public Transport.

<https://www.racv.com.au/cars-transport/bikes-other-transport/public-transport.html#:~:text=Public%20transport%20in%20Victoria,handy%20tips%20on%20using%20myki>